

1 Solution — GIAM 2.2

Problem 2

1.1 Problem

Complete truth tables for the compound sentences $A \Rightarrow B$ and $\neg A \vee B$.

1.2 Answer

The two sentences have **identical** truth tables, so they are logically equivalent:

$$\boxed{A \Rightarrow B \equiv \neg A \vee B.}$$

1.3 The Truth Table

Computing both formulas over all four combinations of A, B (with the helper column $\neg A$):

A	B	$\neg A$	$A \Rightarrow B$	$\neg A \vee B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Table 1.1

The two final columns are identical in all four rows — so $A \Rightarrow B$ and $\neg A \vee B$ are equivalent.

A	B	$\neg A$	$A \Rightarrow B$	$\neg A \vee B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$A \Rightarrow B \equiv \neg A \vee B$

Figure 1.1: A truth table with columns A , B , not- A , A implies B , and not- A or B , over all four rows. Row T,T: not- A is F, both final columns T. Row T,F: not- A F, both final columns F. Row F,T: not- A T, both T. Row F,F: not- A T, both T. The two final columns A implies B and not- A or B are identical in every row, so the boxed conclusion is A implies B is equivalent to not- A or B .

The columns for $A \Rightarrow B$ and $\neg A \vee B$ agree in every row, confirming the equivalence.

1.4 Remark — This Is the *Definition* of Material Implication

The equivalence $A \Rightarrow B \equiv \neg A \vee B$ is often taken as the very **definition** of the logical (material) conditional. It captures the promise “if A , then B ” as “either A fails, or B holds.” Read that way, the only situation that breaks the promise is A true while B false — exactly the single **F** in row 2. Some textbooks introduce $\Rightarrow$ by *stipulating* $\neg A \vee B$ and then deriving its properties; this problem shows the two presentations coincide.

1.5 Remark — Vacuous Truth: Why “False Implies Anything”

The two bottom rows, where A is false, are the ones that puzzle newcomers: $A \Rightarrow B$ comes out **true** whenever A is false, *regardless of* B . The $\neg A \vee B$ form explains why

ous truth: a conditional with a false hypothesis makes no claim that can be violated, so it is counted true by default. “If the moon is made of cheese, then $2 + 2 = 5$ ” is a *true* statement. The equivalence makes this convention feel less arbitrary: it is forced by identifying $\Rightarrow$ with $\neg A \vee B$.

1.6 Remark — A Gateway to the Other Rewrites

Once $A \Rightarrow B \equiv \neg A \vee B$ is in hand, a whole family of identities follows by simple algebra of $\neg, \vee, \wedge$. Negating both sides and applying De Morgan gives the negation of a conditional:

$$\neg(A \Rightarrow B) \equiv \neg(\neg A \vee B) \equiv A \wedge \neg B,$$

“not (if A then B)” means “ A holds but B fails” — again matching row 2, the lone false row. Swapping the disjuncts and re-folding gives the **contrapositive**:

$$\neg A \vee B \equiv B \vee \neg A \equiv \neg(\neg B) \vee (\neg A) \equiv \neg B \Rightarrow \neg A.$$

So $A \Rightarrow B \equiv \neg B \Rightarrow \neg A$ drops out of the same rewrite. The disjunctive form is the hub from which these standard equivalences radiate.

1.7 Remark — Turning Conditionals Into Circuits

The rewrite is also what lets implication be *built* from the basic gates of Problem 2.1.1. There is no “implication gate” among the standard $\{\wedge, \vee, \neg\}$, but $A \Rightarrow B \equiv \neg A \vee B$ says one is unnecessary: feed A through a NOT gate, then OR the result with B . Two gates realise the conditional. This is precisely how conditionals are handled in hardware and in the internals of a SAT solver — every $\Rightarrow$ is silently replaced by its $\neg A \vee B$ form so that the whole formula lives in the common language of $\wedge, \vee, \neg$ (indeed, disjunctive clauses like $\neg A \vee B$ are the native input format — “clauses” — of modern SAT solvers).